

John Doe

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Professional Summary

Mechanical design engineer specializing in test stations, test fixtures, and laboratory systems for aerospace test and evaluation. Takes test equipment from concept through fabrication, integration, and acceptance testing. Background in structural analysis and mechanical testing informs designs that are ready to build and qualify.

Technical Skills

Test & Evaluation: Test Station Design, Test Fixture Design, Acceptance Test Procedures (ATP)

Design Tools: PTC Creo, Siemens NX, AutoCAD

Documentation: Engineering Drawings, GD&T (ASME Y14.5), Bill of Materials (BOM), Windchill PLM

Test Equipment: Data Acquisition (DAQ), Oscilloscopes, Power Supplies, Multimeters, Strain Gauges

Analysis & Manufacturing: ANSYS, Nastran, Design for Manufacturing, Manufacturing Liaison

Professional Experience

Senior Mechanical Design Engineer

August 2020 – Present

Veridane Technical Services (Government Test Center Contract) – Huntsville, AL

- Serve as primary design engineer for a component test laboratory, designing test stations, test fixtures, and laboratory systems from concept through fabrication, integration, and checkout
- Produce solid models, assemblies, and detailed engineering drawings in PTC Creo with GD&T per ASME Y14.5; maintain bills of materials (BOMs) and release through Windchill PLM configuration management
- Led redesign of a legacy valve test station, including its pneumatic and fluid systems per ASME B31.3, achieving a 22% weight reduction
- Developed and executed acceptance test procedures (ATPs) for new and modified test stations
- Integrate instrumentation and data acquisition (DAQ) hardware into test stations, verifying sensor signals with oscilloscopes, power supplies, and multimeters
- Coordinate with a multi-discipline team including supply chain, operations, and quality assurance, and provide manufacturing liaison support during fabrication, assembly, and installation
- Mentor junior engineers on drawing practices and GD&T

Mechanical Engineer – Structural Design & Analysis

June 2017 – July 2020

Korvath Industrial Systems – Huntsville, AL

- Designed welded and bolted structural frames and enclosures, applying GD&T to engineering drawings
- Performed structural analysis of frames and support structure in ANSYS and Nastran; prepared geometry in Siemens NX for meshing
- Resolved recurring fatigue cracking in a production bracket by redesigning the load path and validating the fix with strain gauge testing
- Automated recurring design calculations with Python, reducing analysis turnaround time by 25%

Mechanical Test Technician (Co-op)

May 2015 – May 2017

Pellmont Materials Laboratory – Huntsville, AL

- Executed mechanical testing programs per ASTM standards and designed custom test fixtures, building a working knowledge of test engineering concepts

Education

M.S. Mechanical Engineering, GPA: 3.80

December 2019

Carroway State University

Thesis: Fatigue of Preloaded Bolted Joints

B.S. Mechanical Engineering, GPA: 3.70 (Magna Cum Laude)

May 2017

Carroway State University

Engineering Honor Society, Dean's List

Publications & Presentations

Modular Valve Test Station Design for Leak-Rate Measurement

Aerospace Test & Evaluation Conference, 2024

Thermal Chamber Fixture Design for Sensor Qualification

Regional Test & Measurement Workshop, 2023 (Poster)

Projects

Microcontroller Data Logger | *Personal Project*

Designed and debugged a custom circuit board using an oscilloscope, bench power supply, and logic analyzer

Senior Capstone: Automated Tensile Test Fixture | *First Place, College of Engineering Design Showcase*

Led mechanical design of a fixture that loads and aligns test specimens automatically